Assignment 2- 2: The Planetary Data Policy

1. Introduction & Scenario

Project Starlight's success has attracted new funding! The club has received a special grant for tracking planetary movements, which comes with very specific rules for data retention and cost management that you must implement.

Your assignment is to create a new, dedicated S3 bucket for this planetary data. You will then create and apply a single, comprehensive lifecycle policy to automatically manage the different types of data according to the grant's strict requirements. You must complete this task using only AWS CloudShell.

2. Learning Objectives

This assignment will assess your ability to:

- Independently author a complex, multi-rule S3 lifecycle policy in JSON.
- Use prefixes to apply different lifecycle actions to different datasets within the same bucket.
- Implement both `Transition` (changing storage class) and `Expiration` (deleting) actions.
- Translate a set of business requirements into a functional, programmatic solution.

3. Your Task & Requirements

You must perform the following steps in order within AWS CloudShell.

Step 1: Create and Prepare the Bucket

1. Create a new S3 bucket. It must be globally unique and follow the format `starlight-planetary-yourname-` followed by today's date (e.g., `starlight-planetary-janedoe-20250917`).
2. Enable **versioning** on this bucket immediately after creating it. This is a best practice for data protection and is required by the grant.

Step 2: Prepare the Sample Data

Before creating the policy, you need to create the sample data files that the policy will manage. Run these commands in CloudShell to set up your directory and files.

1. **Create a new directory and move into it:**

   ```
   mkdir planetary_data && cd planetary_data
   ```

2. **Create files for each data type:**

   ```
   # A sample image of Jupiter
   wget https://www.nasa.gov/wp-content/uploads/2024/07/pia25729.jpg -O jupiter-raw.jpg

   # A temporary calibration file
   echo "Telescope calibration run: Temp=2.5C, Focus=Optimal" > daily-calibration.tmp
   ```

```
# A general image of Mars

wget
https://mars.nasa.gov/msl-raw-images/msss/02472/mcam/2472MR013119038070
4379C00_DXXX.jpg -O mars-general.jpg
```

Step 3: Implement the Planetary Data Policy

This is your primary task. You must create a **single JSON file** named `policy.json` that contains a lifecycle configuration with the following three rules:

- **Rule 1 (Jupiter Data):** All objects within the `jupiter-images/` prefix are critical for long-term research. They must be transitioned to the **S3 Glacier Deep Archive** storage class after **180 days**. These files should **never** be deleted by the lifecycle policy.
- **Rule 2 (Calibration Files):** Temporary calibration files, which will be placed in the `calibration-files/` prefix, are only needed for a short time. They must be **permanently deleted 90 days** after they are created.
- **Rule 3 (General Images):** All other general planetary images, which will be placed in the `general-images/` prefix, have unpredictable access patterns. To optimize costs, they should be transitioned to the **S3 Intelligent-Tiering** storage class after **30 days**.

Step 4: Deploy and Apply

1. Upload the three files you created in Step 2 to the S3 bucket, placing them into the correct prefixes as defined by the policy rules (`jupiter-images/`, `calibration-files/`, and `general-images/`).
2. Apply your `policy.json` file to the bucket.

4. Deliverables

Submit a document containing the following two items:

1. **Your S3 Bucket Name:** The full, unique name of the `starlight-planetary-...` bucket you created.
2. **Lifecycle Policy JSON:** The complete and final JSON content of your `policy.json` file. This is the primary component that will be graded.

5. Academic Integrity

This is an individual assignment. You may consult AWS documentation, but the final JSON policy must be authored by you to meet the specific requirements listed above. Your submission must be your own work.